

Novel metabolomic predictors of incident colorectal cancer in men and women

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Abstract

Background: Metabolomic profiles may influence colorectal cancer (CRC) development. Few studies have performed prediagnostic metabolome-wide analyses with CRC risk.

Methods: We conducted a nested case-control study among women (Nurses' Health Study) and men (Health Professionals Follow-Up Study) who provided blood between 1989 and 1995. Over 22.9 years, 684 (409 Nurses' Health Study, 275 Health Professionals Follow-Up Study) individuals developed CRC and were matched 1:1 to unaffected participants. Liquid chromatography-mass spectrometry identified 255 plasma metabolites after quality control. Cohort-specific and combined metabolite association analyses were performed using conditional logistic regression. Metabolite set enrichment analysis was used to identify differential abundance in metabolite classes. The R Weighted Correlation Network Analysis package provided modules of covarying metabolites, which were tested for CRC association.

Results: Metabolite set enrichment analysis identified specific acylcarnitines associated with higher CRC risk and triacylglycerols with lower CRC risk among women and men. Further, phosphatidylcholines were associated with a higher risk of CRC among men. In an analysis restricted to CRC diagnosed 2 years after blood draw, myristoleic acid (odds ratio = 1.37 [95% CI = 1.15 to 1.62]; false discovery rate = 0.072) and C60:12 triacylglycerol (odds ratio = 0.75 [95% CI = 0.64 to 0.88]; false discovery rate = 0.072) were associated with CRC risk in women. Weighted correlation network analysis identified amino acids associated with CRC in men, fatty acid esters (carnitines) with distal CRC in men, and triacylglycerols inversely associated with CRC in women.

Conclusions: We identified prediagnostic CRC-associated metabolites with distinct sex-specific profiles. These results provide insight into CRC etiopathogenesis and have implications for risk prediction strategies.

Introduction

Colorectal cancer (CRC) is the third-most common cancer and the second-leading cause of cancer death in the world.¹ Metabolic disorders are established risk factors for CRC,² suggesting the involvement of metabolic dysregulation in CRC pathogenesis. Recent advances in comprehensive profiling of circulating small

molecules allow for studying metabolite biomarkers related to carcinogenesis. Plasma metabolites reflect dynamic changes in metabolic pathways that can capture the influence of genetic and lifestyle risk factors. For example, branched-chain amino acid plasma levels are associated with type 2 diabetes incidence and have been proposed as early pancreatic cancer biomarkers.^{3,4}

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Most CRC metabolomics studies have used a targeted approach to study a limited number of pathway-specific metabolites. Few prospective studies with long-term follow-up have conducted metabolome-wide analyses for CRC risk, and most have been limited in sample size. In a study of 250 individuals with incident CRC and unaffected individuals within the Shanghai Women's Health Study and the Shanghai Men's Health Study, 9 independent metabolites, including phosphatidylcholines (PCs) and phosphatidylethanolamines (PEs), were associated with CRC risk.⁵ In the Prostate, Lung, Colorectal, and Ovarian Cancer study, no metabolites were associated with CRC among 254 patients with CRC and 254 matched control individuals.⁶ An untargeted study of 902 patients with CRC and matched control individuals from the Northern Sweden Health and Disease Study⁷ found only 1 unknown metabolite, at a false discovery rate below 0.10, likely due to limited statistical power from the high number of unknown tested metabolites. Therefore, large-scale, untargeted approaches testing the association of known metabolites with CRC risk from prospective cohorts are imperative. We examined the association of metabolomic profiles with CRC incidence among men and women in 2 large, prospective US-based cohorts.

Methods

Study population

We conducted nested case-control analyses within 2 US-based cohorts: the Nurses' Health Study (NHS), a prospective cohort started in 1976 of 121 700 female nurses, and the Health Professionals Follow-Up Study (HPFS), a prospective cohort of 51 529 male health professionals started in 1986. Incident diagnoses and health-related information were obtained from biennial health questionnaires. In addition, death information was captured from the National Death Index and next-of-kin confirmation.⁸ In 1989-1990, 32 826 NHS participants and in 1993-1995, 18 225 HPFS participants provided blood samples, as previously described.^{9,10} The study protocol was approved by the institutional review boards of the Brigham and Women's Hospital and Harvard T.H. Chan School of Public Health as well as those of participating registries, as required.

CRC ascertainment and nested case-control study design

Participant reports of CRC were adjudicated by medical record review. Information regarding cancer stage, histological subtype, and tumor location was recorded. Eligible cases were defined as individuals who provided blood specimens with subsequent diagnoses of CRC until the 2010 follow-up cycles in both cohorts. Incident CRC diagnoses were matched 1:1 to unaffected individuals based on sex, age, month of blood collection, and self-reported fasting status (last meal >8 hours) at blood collection.

Metabolite assessment

Plasma metabolites were profiled at the Broad Institute (Cambridge, MA) using liquid chromatography-tandem mass spectrometry. Specifically, metabolites were profiled using a liquid chromatography-mass spectrometry platform consisting of a Shimadzu Nexera X2 Ultra-High-Performance Liquid Chromatography coupled to a Q Exactive mass spectrometer (Thermo Fisher Scientific). Hydrophilic interaction liquid chromatography was used to profile water-soluble metabolites (amino acids, dipeptides, and other cationic molecules) in the positive ionization mode. Polar and nonpolar lipids were assessed using the C8-positive approach. These methods are described in

detail elsewhere.¹⁰⁻¹³ More than 5000 metabolites were measured. Our study was restricted to the 362 metabolites that could be identified and annotated by the Human Metabolome Database. Nine metabolites sensitive to delayed processing in a prior study were excluded.¹³ Metabolites with a coefficient of variation (s/x) $\times 100$ greater than 75% or an intraclass correlation coefficient less than 0.75 among blinded quality control samples were excluded, which resulted in the removal of 50 metabolites.

Statistical analysis

SAS, version 9, statistical software (SAS Institute Inc) was used to obtain all participant-related data. All statistical analyses were performed within an R, version 4.3.2, statistical programming environment (R Foundation for Statistical Computing).

Metabolite values were log-transformed and standardized to a mean (SD) value of zero (1). Metabolites not reported in more than 10% of cohort participants were removed, which resulted in the removal of an additional 48 metabolites. The remaining missing metabolite values were imputed by one-half the minimum value for the respective metabolite in a cohort-specific manner. In total, 255 metabolites were tested for their association with incident CRC. The *clogit* function within the R *survival*, version 3.5-7, package was used to perform conditional logistic regression to obtain single-metabolite CRC odds ratios (ORs) and 95% CIs separately in the NHS, HPFS, and combined cohorts. Metabolites were modeled as continuous variables. Univariate conditional models accounted for matching factors. Multivariate conditional models were further adjusted for CRC family history, body mass index (BMI), smoking history (none, <40 pack-years, >40 pack-years), alcohol consumption (g/d), Alternative Healthy Eating Index,¹⁴ physical activity (metabolic equivalent-hours/week), regular aspirin use (≥ 2 times per week) as aspirin is protective against CRC,¹⁵ menopausal/hormone use status (premenopausal, postmenopausal/no hormone use, postmenopausal/past hormone use, postmenopausal/current hormone use). In addition, sex was controlled for in the combined cohort models. Individuals identifying as any race/ethnicity other than Non-Hispanic White were categorized as 'Other.' This information was collected to assess potential confounding but was not incorporated into the multivariable model. These covariates were calculated as the cumulative average from all biennial questionnaires before blood donation. Missing continuous covariates were median imputed and missing categorical covariates with the most frequent category value. For associations of individual metabolites, we corrected multiple comparisons using the Benjamini-Hochberg procedure, with a false discovery rate less than 0.10.

Subgroup analyses

To test whether specific metabolites were associated with CRC tumor location, we defined *proximal* CRC as tumors found in the cecum, ascending, and transverse colon; *distal* CRC as tumors found in the descending colon and sigmoid colon; and *rectal cancer* as tumors found in the rectosigmoid junction and rectum.

Metabolite set enrichment analysis

We identified metabolite class abundance differences between affected and unaffected individuals in a metabolite set enrichment analysis.¹⁶ To perform metabolite set enrichment analysis, the odds ratio of each tested metabolite was log-transformed and sorted in descending order. The *ClusterProfiler*,¹⁷ version 4.10.0, R package was used to detect statistically significant (false

discovery rate < 0.10) metabolite Human Metabolome Database class-type enrichments.

Weighted Correlation Network Analysis

The *Weighted Correlation Network Analysis* (WGCNA), version 1.72-5, R package¹⁸ was used to identify and quantify modules of covarying metabolites in each participant. Specifically, hierarchical clustering of metabolite covariance from HPFS and NHS control individuals was performed to identify cohort-specific “signed” network modules, which are arbitrarily assigned a unique color label. The formation of these modules was not prespecified by any known biological characteristics. Modules were optimized to achieve scale-free topology by selecting a cohort-specific optimal soft thresholding power. The minimum number of metabolites allowed to form a module was 10. A dynamic tree cut method merged closely related metabolite modules.¹⁹ Module “eigenmetabolite” score, the first principal component of the respective module, was calculated to quantify each module's relative abundance for each participant. These eigenmetabolite scores were z-transformed and tested in a univariate and multivariable conditional logistic regression to test for CRC and CRC location subtype association. A consensus network was constructed to identify shared HPFS and NHS modules and used in a combined conditional regression analysis to identify non-sex-specific metabolite networks associated with CRC. Overrepresentation analysis was performed using MetaboAnalyst, version 6.0, software to identify whether modules were composed of specific metabolite subclasses or Relational database of Metabolomic Pathways biological pathways, at a false discovery rate of 0.10.^{20,21}

Results

Cohort characteristics

Over the follow-up period, 684 individuals developed CRC (409 in NHS; 275 in HPFS) after blood draw. The median time from blood collection to CRC diagnosis was 128 months (10.7 years) in the NHS cohort and 81 months (6.8 years) in the HPFS cohort. These participants with CRC were individually matched to 684 unaffected individuals, with a median age at blood draw of 60 years in NHS and 68 years HPFS (Table 1). In total, 68% of NHS and 58% of HPFS participants provided fasting blood samples. In the HPFS nested case-control study, cases had a significantly higher BMI (Table 1). Otherwise, the matched case-control cohorts were well balanced.

Metabolites associated with the risk of CRC

No metabolite was associated with CRC risk in the female or combined cohorts, at an false discovery rate threshold below 0.10 (Table 2 and Table S1). Six metabolites, however, were associated with CRC risk in the female nested case-control cohort, and 1 metabolite was associated with CRC in the male cohort, at an false discovery rate of 0.25 (Table 2 and Table S1), a threshold previously used to define significant CRC-associated metabolites.⁷ A nucleoside molecule, N2, N2-dimethylguanosine, was associated with CRC risk (OR = 1.55 [95% CI = 1.22 to 1.96]; false discovery rate = 0.084) in men in the univariate model; however, this association was attenuated in the multivariate model (OR = 1.53 [95% CI = 1.19 to 1.95]; false discovery rate = 0.191) (Table 2 and Table S1).

Metabolite set enrichment analysis found an enrichment of triacylglycerols among metabolites that were inversely associated with CRC and a significant enrichment of acylcarnitines among metabolites associated with CRC risk in the male, female, and combined cohorts (Figure 1, A, Table S2). In addition,

enrichment of phosphatidylcholines was found among metabolites associated with CRC in men.

In a lag analysis in which we removed CRC diagnoses within 2 years of blood collection to account for potential prevalent CRC, 2 metabolites—myristoleic acid (increased risk) and C60:12 TAG (triacylglycerol; reduced risk)—were found to be associated with CRC risk in women (Table 3). No significant associations were observed in the male or the combined cohorts in lag analyses (Table S3). No single metabolite was associated with proximal colon, distal colon, or rectal cancer, but 5 acylcarnitines (C10 carnitine, C8 carnitine, C12 carnitine, C12:1 carnitine, and C14:2 carnitine) were associated with increased distal colon cancer risk in the combined cohort univariate model (Table S4).

Association of metabolite classes with CRC risk

WGCNA found 7 metabolite modules in NHS, 9 in HPFS, and 8 in the consensus network (Figures S1-S5). The *Light Salmon* NHS module was inversely associated with CRC (OR = 0.78 [95% CI = 0.67 to 0.92]; false discovery rate = 0.016) (Table 4). This module consisted of 27 metabolites and was enriched for triacylglycerols (false discovery rate = 0.0011). Seventeen metabolites in this module were associated with reduced CRC risk at nominal significance ($P < .05$), with the strongest association observed for C60:12 TAG (Figure 1, B). The *Light Salmon* NHS module metabolites belonged to 3 different modules in the HPFS network (*Dark Magenta*, *Dark Orchid*, and *Dark Slate Blue*), none of which was associated with CRC in men.

A single HPFS module (*Dark Salmon*) was associated with CRC risk in men (OR = 1.40 [95% CI = 1.10 to 1.79]; false discovery rate = 0.058) (Table 4). This module consisted of 28 metabolites, 8 of which were nominally associated with increased CRC risk. This module was enriched for “amino acids, peptides, and analogues” (false discovery rate = $1.27E^{-4}$). N2, N2-dimethylguanosine had the strongest association with CRC risk within this module (Figure 1, B). No consensus modules were associated with CRC in the combined cohort.

The WGCNA modules were also tested for their association with CRC location. The *Dark Salmon* HPFS module and the *Dark Orange* module were associated with distal colon cancer in men (OR = 1.93 [95% CI = 1.21 to 3.06]; false discovery rate = 0.050) (Table 4). The *Dark Orange* module consists of 18 metabolites, 12 of which were carnitines, and was enriched for “fatty acid esters” (false discovery rate = $7.50E^{-10}$). Nine metabolites, all carnitines, were nominally associated with distal colon cancer in the male single-metabolite analysis. The *Black* module in the consensus WGCNA network, which consisted of 12 carnitines also found in the HPFS *Dark Orange* module, was associated with distal colon cancer in the combined cohort. These carnitines were also found in the NHS *Light Yellow* module in the NHS WGCNA network; however, this module was not associated with CRC risk in women. No associations were found with proximal colon cancer or rectal cancer in men or women. Table S5 contains the list of metabolites and overrepresentation analysis results of each module.

Discussion

We used untargeted metabolomics in prediagnosis plasma samples from 684 matched CRC case-control pairs across 2 large, prospective, US-based cohorts. We observed some variations in metabolomic profiles between women and men with CRC that could indicate sex-specific differences in CRC pathophysiology, consistent with prior studies. For instance, 1 investigation

Table 1. Characteristics of individuals with incident CRC and matched control individuals in women (NHS) and men (HPFS).

Baseline characteristic	NHS (women)			HPFS (men)		
	Control individuals (n = 409)	Patients (n = 409)	P	Control individuals (n = 275)	Patients (n = 275)	P
Age at blood draw, median (IQR), y	60 (54-65)	60 (54-64)	>.9 ^a	68 (60-72)	68 (60-72)	>.9 ^a
Age at CRC diagnosis, median (IQR), y	—	70 (64-75)	—	—	74 (67-80)	—
Time from blood draw to CRC diagnosis, median (IQR), mo	—	128 (74-186)	—	—	81 (43-115)	—
Fasting at blood draw, No. (%)	280 (68)	279 (68)	>.9 ^b	162 (59)	157 (57)	.7 ^b
CRC location, No. (%)			—			—
Proximal	—	201 (51)		—	104 (42)	
Distal	—	109 (28)		—	75 (31)	
Rectal	—	85 (22)		—	66 (27)	
Family history of CRC, No. (%)	55 (13)	60 (15)	.6 ^b	42 (15)	56 (20)	.12 ^b
Non-Hispanic White race, No. (%)			.2 ^c			.045 ^c
Other	7 (1.7)	13 (3.2)		8 (2.9)	18 (6.5)	
Non-Hispanic White	402 (98)	396 (97)		267 (97)	257 (93)	
Postmenopausal/hormone use status, No. (%)			.5 ^b			—
Premenopausal	60 (15)	62 (15)		—	—	
Postmenopausal, no hormone use	131 (32)	124 (30)		—	—	
Postmenopausal, past hormone use	56 (14)	72 (18)		—	—	
Postmenopausal, current hormone use	162 (40)	151 (37)		—	—	
Smoking history, No. (%)			.8 ^b			.5 ^b
Never	181 (44)	183 (45)		138 (50)	129 (47)	
<40 pack-y	186 (45)	179 (44)		115 (42)	116 (42)	
>40 pack-y	42 (10)	47 (11)		22 (8.0)	30 (11)	
Body mass index (kg/m ²), median (IQR)	23.7 (21.7-26.3)	23.9 (21.9-26.9)	.3 ^a	24.9 (23.4-26.8)	25.9 (24.2-27.5)	<.001 ^a
Physical activity (metabolic equivalent-h/wk), median (IQR)	10 (4-21)	10 (4-19)	.7 ^a	24 (12-35)	23 (11-38)	.8 ^a
Alternate Healthy Eating Index, median (IQR)	46 (41-53)	46 (39-52)	.4 ^a	48 (40-55)	48 (41-55)	.8 ^a
Alcohol, median (IQR), g/d	2 (0-10)	2 (1-9)	.9 ^a	7 (2-16)	8 (1-19)	.7 ^a
Regular aspirin user (≥2 tabs/wk), No. (%)	183 (45)	147 (36)	.01 ^b	131 (48)	126 (46)	.7 ^b

Abbreviations: CRC = colorectal cancer; HPFS = Health Professionals Follow-Up Study; NHS = Nurses' Health Study.

^a Wilcoxon rank sum test.^b Pearson χ^2 test.^c Fisher exact test.

reported that genetically influenced BMI is associated with CRC risk in women but not men.²² Several genetic variants that influenced BMI in this study were found in or near genes involved in lipid and acylglycerol metabolism.^{22,23} Consistent with these findings, we observed stronger associations for lipids/acylglycerol metabolites among women than men. A recent analysis of the participants in this study found differences in the association of inflammatory biomarkers with CRC,²⁴ further reinforcing sex-dependent metabolite profiles associated with CRC. The apparent sex differences in metabolite profiles could be due to differences in statistical power, however, given that metabolite set enrichment analysis identified overlapping classes of risk metabolites. Furthermore, confounding variables not adjusted for in our multivariable models could account for the differences in our findings. Additional studies to replicate our findings are therefore warranted.

We found that myristoleic acid was positively associated with CRC risk in women. Myristoleic acid is an omega-5 monounsaturated fatty acid that induces apoptosis in prostate cancer cells.²⁵ Myristoleic acid also induces Wnt/ β -catenin signaling, a disordered signaling pathway in CRC and cell cycle progression.²⁶ Dietary intake of myristoleic acid,²⁷ serum myristic acid,²⁸ and serum myristoleic acid are associated with incident breast cancer.¹² Given our findings, myristoleic acid may serve as a novel multicancer risk factor or a marker for dietary risk factors. Myristoleic acid is also synthesized through the de novo lipogenesis pathway. Given that normal de novo lipogenesis is altered in

obesity and metabolic diseases,^{29,30} this could indicate that myristoleic acid is a marker of disordered metabolism, a CRC risk factor.³¹ Future studies are needed to identify the sources of myristoleic acid and how it translates to cancer risk.

Among men, we observed that N2, N2-dimethylguanosine was associated with CRC in a univariate model and nominally significant in a multivariate model. Further, it was the most significant CRC-associated metabolite in the WGCNA *Dark Salmon* module in men. N2, N2-dimethylguanosine is a nucleoside and degradation product of transfer RNAs.³² Urinary N2, N2-methylguanosine is elevated in patients with CRC,³² and serum concentrations of N2, N2-dimethylguanosine have been associated with CRC disease progression.³³ This process may be due to the increased activity of transfer RNA methyltransferase in cancer tissues³⁴ and rapid turnover of cancer tissue transfer RNA,³⁵ suggesting a role for N2, N2-dimethylguanosine as a potential early diagnosis marker.³² Moreover, a metabolome-wide association study reported an association for N2, N2-dimethylguanosine with pancreatic cancer risk, indicating involvement in multiple gastrointestinal cancers.³⁶

We observed an association of carnitine metabolites with increased distal colon cancer risk in men using WGCNA. Carnitines facilitate the movement of acetyl-CoA into mitochondria and may affect carcinogenesis through several mechanisms, including inflammation and apoptosis.³⁷⁻³⁹ Several carnitines have been reported to be elevated in CRC patient serum, urine, and stool.³⁸ The effect size of carnitines was more substantial

Table 2. Association of individual metabolites with risk of CRC in women and men after adjusting for CRC risk factors

Cohort	Human Metabolome Database identifier	Metabolite	Method	NHS ^a			HPFS ^b		
				Multivariable OR (95% CI) ^c	Multivariable p ^d	Multivariable false discovery rate	Multivariable OR (95% CI) ^c	Multivariable p ^d	Multivariable false discovery rate
NHS	HMDB0005478 ^e	C60:12 TAG	C8-pos	0.77 (0.66 to 0.90)	9.19 × 10 ⁻⁴	0.128	1.05 (0.8 to 1.27)	.641	0.972
	HMDB0010531 ^e	C58:11 TAG	C8-pos	0.77 (0.66 to 0.90)	.001	0.128	1.01 (0.83 to 1.23)	.911	0.979
	HMDB0002000	Myristoleic acid	HILIC-pos	1.30 (1.11 to 1.53)	.002	0.128	0.85 (0.69 to 1.05)	.124	0.807
	HMDB0005463 ^e	C58:9 TAG	C8-pos	0.81 (0.69 to 0.94)	.005	0.232	1.01 (0.83 to 1.24)	.895	0.979
	HMDB0005448 ^e	C56:9 TAG	C8-pos	0.80 (0.69 to 0.94)	.005	0.232	0.98 (0.80 to 1.18)	.804	0.979
	HMDB0008731 ^e	C40:9 PC	C8-pos	0.80 (0.69 to 0.94)	.005	0.232	1.06 (0.86 to 1.30)	.586	0.955
	HMDB0008925 ^e	C34:0 PE	C8-pos	0.81 (0.69 to 0.94)	.007	0.246	1.09 (0.91 to 1.31)	.333	0.955
	HMDB0007991 ^e	C38:6 PC	C8-pos	0.81 (0.69 to 0.95)	.008	0.246	1.09 (0.88 to 1.34)	.431	0.955
	HMDB0005462 ^e	C56:7 TAG	C8-pos	0.82 (0.71 to 0.96)	.011	0.285	0.94 (0.78 to 1.13)	.493	0.955
	HMDB0013631	Oleoyl glycine	HILIC-pos	1.23 (1.05 to 1.45)	.011	0.285	1.09 (0.89 to 1.33)	.431	0.955
	HMDB0008057 ^e	C40:6 PC	C8-pos	0.83 (0.72 to 0.97)	.015	0.320	1.08 (0.89 to 1.31)	.453	0.955
	HMDB0013733 ^e	Trimethylbenzene	HILIC-pos	1.21 (1.04 to 1.41)	.015	0.320	1.10 (0.90 to 1.35)	.331	0.955
	HMDB0010498 ^e	C54:9 TAG	C8-pos	0.83 (0.71 to 0.97)	.016	0.320	1.01 (0.84 to 1.21)	.920	0.979
	HMDB0005392 ^e	C56:8 TAG	C8-pos	0.84 (0.72 to 0.97)	.018	0.322	0.98 (0.81 to 1.18)	.802	0.979
	HMDB0000025	Thiamine	HILIC-pos	0.83 (0.72 to 0.97)	.022	0.322	0.97 (0.80 to 1.16)	.724	0.979
	HMDB0011526	C22:6 LPE	C8-pos	0.84 (0.72 to 0.97)	.022	0.322	1.12 (0.91 to 1.36)	.279	0.955
	HMDB0004827	Proline betaine	HILIC-pos	0.84 (0.72 to 0.98)	.023	0.322	1.06 (0.89 to 1.26)	.523	0.955
	HMDB0000138	Glycocholate	HILIC-pos	1.18 (1.02 to 1.36)	.025	0.322	1.07 (0.89 to 1.29)	.474	0.955
	HMDB0011420 ^e	C38:7 PE plasmalogen	C8-pos	0.85 (0.74 to 0.98)	.026	0.322	1.07 (0.88 to 1.30)	.526	0.955
	HMDB0000064	Creatine	HILIC-pos	0.85 (0.73 to 0.98)	.028	0.322	1.09 (0.89 to 1.32)	.401	0.955
	HMDB0013288	C9 carnitine	HILIC-pos	0.84 (0.72 to 0.98)	.029	0.322	1.15 (0.94 to 1.40)	.182	0.910
	HMDB0010518 ^e	C54:8 TAG	C8-pos	0.85 (0.73 to 0.98)	.029	0.322	0.94 (0.78 to 1.13)	.505	0.955
	HMDB0005456 ^e	C56:6 TAG	C8-pos	0.85 (0.73 to 0.98)	.029	0.322	0.98 (0.82 to 1.18)	.862	0.979
	HMDB0000182	Lysine	HILIC-pos	0.84 (0.72 to 0.99)	.034	0.362	0.93 (0.76 to 1.12)	.437	0.955
	HMDB0002820	Methylimidazole acetic acid	HILIC-pos	0.85 (0.74 to 0.99)	.037	0.374	1.16 (0.96 to 1.40)	.118	0.793
	HMDB00000201	C2 carnitine	HILIC-pos	1.18 (1.01 to 1.39)	.039	0.386	1.20 (0.99 to 1.44)	.065	0.733
	HMDB0007869 ^e	C30:0 PC	C8-pos	0.85 (0.73 to 1.00)	.044	0.405	1.10 (0.90 to 1.33)	.352	0.955
	HMDB0012356 ^e	C34:0 PS	C8-pos	0.86 (0.74 to 1.00)	.045	0.405	1.09 (0.89 to 1.33)	.423	0.955
	HMDB0010517 ^e	C52:7 TAG	C8-pos	0.86 (0.74 to 1.00)	.046	0.405	0.96 (0.80 to 1.16)	.693	0.979

(continued)

Table 2. (continued)

Cohort	Human Metabolome Database identifier	Metabolite	Method	NHS ^a			HPFS ^b		
				Multivariable OR (95% CI) ^c	Multivariable p ^d	Multivariable false discovery rate	Multivariable OR (95% CI) ^c	Multivariable p ^d	Multivariable false discovery rate
HPFS	HMDB0004824	N2, N2-dimethylguanosine	HILIC-pos	1.02 (0.85 to 1.21)	.865	0.998	1.53 (1.19 to 1.95)	7.47 × 10 ⁻⁴	.191
	HMDB0000248	Thyroxine	HILIC-pos	0.99 (0.85 to 1.15)	.894	0.999	1.42 (1.12 to 1.81)	.004	0.340
	HMDB0006344	Phenylacetylglutamine	HILIC-pos	0.99 (0.85 to 1.15)	.896	0.999	1.32 (1.08 to 1.60)	.005	0.340
	HMDB0000026	N-carbamoyl-beta-alanine	HILIC-pos	1.00 (0.87 to 1.15)	.993	0.999	1.33 (1.09 to 1.62)	.006	0.340
	HMDB0000092	Dimethylglycine	HILIC-pos	1.01 (0.88 to 1.17)	.847	0.995	1.31 (1.07 to 1.59)	.009	0.340
	HMDB0000766	N-acetyalanine	HILIC-pos	0.96 (0.82 to 1.13)	.617	0.906	1.38 (1.08 to 1.76)	.009	0.340
	HMDB0004620	N-alpha-acetylarginine	HILIC-pos	0.89 (0.77 to 1.04)	.148	0.541	1.29 (1.06 to 1.56)	.009	0.340
	HMDB0000824	C3 carnitine	HILIC-pos	0.96 (0.83 to 1.11)	.592	0.891	1.29 (1.06 to 1.57)	.013	0.367
	HMDB0000544	5-hydroxymethyl-4-methyluracil	HILIC-pos	0.89 (0.76 to 1.03)	.119	0.518	1.32 (1.06 to 1.65)	.013	0.367
	HMDB0011214 ^e	C34:5 PC plasmalogen	C8-pos	0.92 (0.80 to 1.05)	.199	0.619	0.78 (0.64 to 0.95)	.016	0.404
	HMDB0004949	C16:0 Ceramide (d18:1)	C8-pos	1.07 (0.92 to 1.24)	.384	0.782	1.24 (1.02 to 1.51)	.028	0.640
	HMDB0000177	Histidine	HILIC-pos	0.93 (0.78 to 1.10)	.373	0.782	0.81 (0.66 to 0.99)	.036	0.690
	HMDB0000767	Pseudouridine	HILIC-pos	0.93 (0.79 to 1.09)	.359	0.782	1.28 (1.01 to 1.61)	.040	0.690
	HMDB0013678	4-Hydroxyhippurate	HILIC-pos	0.96 (0.83 to 1.11)	.552	0.871	1.21 (1.01 to 1.45)	.041	0.690
	HMDB0002869	Campesterol	C8-pos	0.99 (0.86 to 1.13)	.837	0.993	1.25 (1.01 to 1.54)	.041	0.690
	HMDB0008038 ^e	C36:1 PC	C8-pos	1.00 (0.87 to 1.16)	.950	0.999	1.23 (1.00 to 1.51)	.047	0.733

Abbreviations: CI = confidence interval; CRC = colorectal cancer; HILIC = hydrophilic interaction liquid chromatography used to profile water-soluble metabolites in the positive ionization mode; HMDB = Human Metabolome Database; HPFS = Health Professionals Follow-Up Study; LPE = lysophosphatidylethanolamine; NHS = Nurses Health Study; PC = phosphatidylcholine; PE = phosphatidylethanolamine; OR = odds ratio; TAG = triacylglycerol; UV = univariable.

^a 409 matched case-control pairs.

^b 275 matched case-control pairs.

^c Conditional odds ratios adjusted for family history of CRC, body mass index, smoking history, alcohol consumption (g/d), Alternative healthy Eating Index, physical activity, menopausal/hormone use status, and regular aspirin use in addition to already-matched variables, odds ratio association for 1 SD increase in log-metabolite values.

^d Only the metabolite associations with P ≤ .05 in either cohort are shown.

^e Representative Human Metabolome Database identifier.

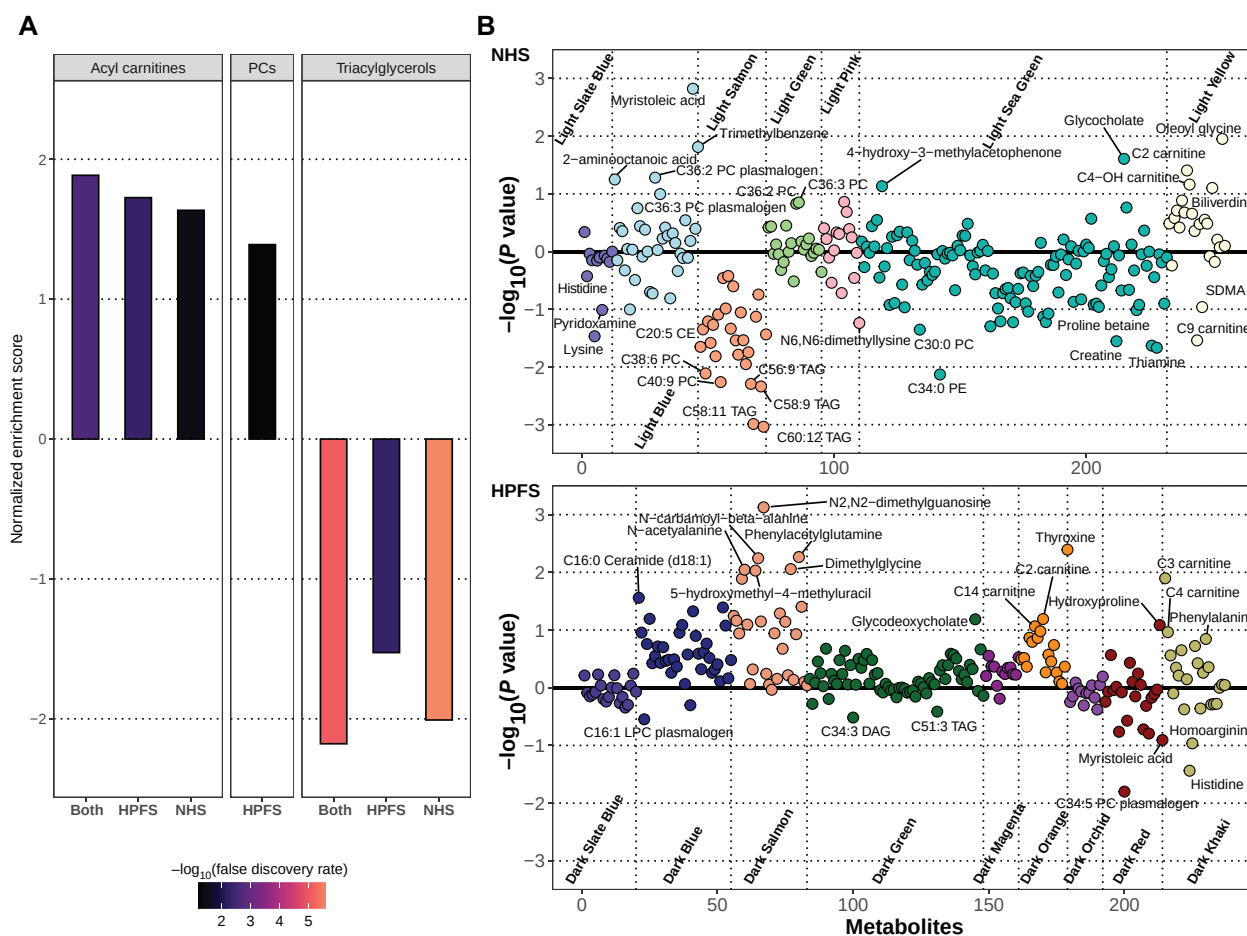


Figure 1. Gene set enrichment analysis and the *Weighted Correlation Network Analysis* package identify metabolic signatures associated with CRC risk. **A)** Enriched metabolite set enrichment analysis terms, by cohort, with an false discovery rate <0.10. Positive normalized enrichment scores indicate enrichment among positively associated CRC metabolites and the opposite for negative normalized enrichment scores. **B)** Miami plot showing the $-\log_{10}(\text{P value})$ association of all metabolites, grouped into modules, with CRC risk by nested case-control cohort. The strongest NHS modular association was the *Light Salmon* module, which was enriched for triacylglycerols, with C60:12 TAG being the most strongly associated CRC metabolite within that module. The HPFS *Dark Salmon* module, enriched for peptides, had the strongest association with CRC, with N2, N2-dimethylguanosine being the most CRC-associated metabolite within that module. CRC = colorectal cancer; HPFS = Health Professionals Follow-Up Study; NHS = Nurses' Health Study; PC = phosphatidylcholine; PE = phosphatidylethanolamine; TAG = triacylglycerol.

with distal colon cancer than all CRC or other location subtypes, which is consistent with previous findings showing that metabolic risk factors are associated with distal CRC but not proximal CRC.³¹

Many previously reported CRC-associated metabolites were not replicated in our study. A targeted metabolite study found plasma choline metabolites associated with postmenopausal CRC risk.⁴⁰ Further, there was a weak inverse association between choline and betaine with CRC risk in the European Prospective Investigation into Cancer (EPIC) cohort.⁴¹ In the same cohort, 7 of 17 measured conjugated primary and secondary bile acids were associated with colon cancer.⁴² We could not comprehensively validate previous associations of these metabolites because our liquid chromatography-mass spectrometry platforms did not capture bile acids well; however, glycocholate was associated with CRC risk in women at nominal significance (OR=1.18 [95% CI=1.02 to 1.36]). A metabolomic signature of odd-chain fatty acids, serine, glycine, and phosphatidylcholines with a healthy diet and lifestyle was associated with reduced CRC risk in the EPIC cohort.⁴³ Recently, targeted quantification of amino acids found that histidine levels were inversely associated with CRC in the EPIC and UK Biobank cohorts.⁴⁴ Although many of these

metabolites were measured in our metabolome-wide study, their associations were insignificant after multiple test correction.

Compared with prior untargeted metabolomics studies, we did not confirm the findings from an Asian cohort that specific glycerophospholipids, including phosphatidylcholines and phosphatidylethanolamines, were inversely associated with CRC risk.⁵ Other phosphatidylcholines, however, were associated with CRC in men in our metabolite set enrichment analysis. Further, we could not validate the metabolite associations found in the Northern Sweden Health and Disease Study CRC study.⁷ Given the limited power to detect all true associations, our lack of replication does not invalidate these reported metabolites as markers of carcinogenesis.

The strengths of our study include a large sample size to detect novel associations and a prospective study design with approximately 2 decades of follow-up, minimizing the likelihood of reverse causation. In addition, we used stringent quality and significance thresholds to avoid spurious findings. Finally, comprehensive medical, dietary, and lifestyle factors collected with a high follow-up rate enabled us to account for many known CRC risk factors. Our study has several limitations, as well. First, metabolites were measured at a single time point, which

Table 3. Association of individual metabolites with risk of CRC in women after adjusting for CRC risk factors and removing CRC diagnosed within 2 years of blood draw

Human Metabolome Database identifier	Metabolite	Method	NHS ^a			HPFS ^b		
			Multivariable OR (95% CI) ^c	Multivariable P ^d	Multivariable false discovery rate	Multivariable OR (95% CI) ^c	Multivariable P ^d	Multivariable false discovery rate
HMDB0002000	Myristoleic acid	HILIC-pos	1.37 (1.15 to 1.62)	3.93 × 10 ⁻⁴	0.072 ^e	0.91 (0.72 to 1.15)	.429	0.865
HMDB0005478 ^f	C60:12 TAG	C8-pos	0.75 (0.64 to 0.88)	5.62 × 10 ⁻⁴	0.072 ^e	1.00 (0.81 to 1.24)	.982	0.993
HMDB0010531 ^f	C58:11 TAG	C8-pos	0.77 (0.65 to 0.90)	.001	.0104	1.00 (0.81 to 1.24)	.964	0.993
HMDB0013631	Oleoyl glycine	HILIC-pos	1.29 (1.09 to 1.53)	.003	.0181	1.07 (0.85 to 1.35)	.559	0.909
HMDB0005463 ^f	C58:9 TAG	C8-pos	0.79 (0.68 to 0.93)	.004	.0181	1.02 (0.82 to 1.26)	.883	0.954
HMDB0008731 ^f	C40:9 PC	C8-pos	0.80 (0.68 to 0.94)	.006	.0211	1.06 (0.85 to 1.33)	.607	0.917
HMDB0005448 ^f	C56:9 TAG	C8-pos	0.80 (0.68 to 0.94)	.007	.0211	1.01 (0.82 to 1.24)	.943	0.990
HMDB0013733 ^f	Trimethylbenzene	HILIC-pos	1.25 (1.06 to 1.47)	.007	.0211	1.16 (0.92 to 1.45)	.206	0.717
HMDB0007991 ^f	C38:6 PC	C8-pos	0.80 (0.68 to 0.94)	.007	.0211	1.08 (0.86 to 1.36)	.502	0.891
HMDB0008925 ^f	C34:0 PE	C8-pos	0.80 (0.68 to 0.95)	.009	.0220	1.06 (0.87 to 1.29)	.567	0.909
HMDB0000235	Thiamine	HILIC-pos	0.81 (0.69 to 0.95)	.012	.0247	1.05 (0.86 to 1.29)	.611	0.917
HMDB0011420 ^f	C38:7 PE plasmalogen	C8-pos	0.83 (0.71 to 0.96)	.012	.0247	1.07 (0.86 to 1.33)	.536	0.899
HMDB0000064	Creatine	HILIC-pos	0.82 (0.70 to 0.96)	.014	.0250	1.19 (0.95 to 1.48)	.122	0.708
HMDB0005462 ^f	C56:7 TAG	C8-pos	0.82 (0.70 to 0.96)	.014	.0250	1.00 (0.82 to 1.23)	.978	0.993
HMDB0005392 ^f	C56:8 TAG	C8-pos	0.82 (0.70 to 0.96)	.015	.0250	1.02 (0.82 to 1.25)	.883	0.954
HMDB0008057 ^f	C40:6 PC	C8-pos	0.83 (0.71 to 0.97)	.016	.0260	1.07 (0.87 to 1.32)	.503	0.891
HMDB0013288	C9 Carnitine	HILIC-pos	0.82 (0.70 to 0.97)	.019	.0283	1.16 (0.94 to 1.45)	.173	0.717
HMDB0011526	C22:6 LPE	C8-pos	0.83 (0.71 to 0.97)	.023	.0327	1.06 (0.85 to 1.32)	.580	0.916
HMDB0000182	Lysine	HILIC-pos	0.83 (0.70 to 0.98)	.028	.0360	0.95 (0.77 to 1.19)	.680	0.919
HMDB0010498 ^f	C54:9 TAG	C8-pos	0.84 (0.72 to 0.98)	.030	.360	1.02 (0.84 to 1.24)	.848	0.954
HMDB0011394 ^f	C40:7 PE plasmalogen	C8-pos	0.85 (0.74 to 0.99)	.032	.360	1.07 (0.86 to 1.32)	.563	0.909
HMDB0000138	Glycocholate	HILIC-pos	1.18 (1.01 top 1.37)	.034	.360	1.15 (0.93 to 1.41)	.191	0.717
HMDB0005456 ^f	C56:6 TAG	C8-pos	0.85 (0.72 to 0.99)	.034	.360	0.98 (0.80 to 1.19)	.827	0.954
HMDB0002820	Methylimidazole acetic acid	HILIC-pos	0.85 (0.72 to 0.99)	.035	.360	1.15 (0.94 to 1.41)	.179	0.717
HMDB0011229 ^f	C38:7 PC plasmalogen	C8-pos	0.85 (0.73 to 0.99)	.035	.360	0.94 (0.76 to 1.17)	.566	0.909
HMDB0009102 ^f	C38:6 PE	C8-pos	0.84 (0.71 to 0.99)	.040	.368	1.06 (0.85 to 1.33)	.607	0.917
HMDB0003334	SDMA	HILIC-pos	0.83 (0.70 to 0.99)	.040	.368	1.17 (0.93 to 1.46)	.186	0.717
HMDB0002005	Methionine sulfoxide	HILIC-pos	0.84 (0.71 to 0.99)	.042	.368	1.01 (0.80 to 1.27)	.955	0.993
HMDB0002366	C5:1 Carnitine	HILIC-pos	0.85 (0.73 to 0.99)	.042	.368	0.98 (0.80 to 1.20)	.838	0.954
HMDB0042104 ^f	C51:1 TAG	C8-pos	0.85 (0.72 to 1.00)	.049	.392	1.02 (0.83 to 1.25)	.865	0.954
HMDB0007869 ^f	C30:0 PC	C8-pos	0.85 (0.72 to 1.00)	.050	.392	1.03 (0.83 to 1.28)	.798	0.954

Abbreviations: CRC = colorectal cancer; HILIC = hydrophilic interaction liquid chromatography used to profile water-soluble metabolites in the positive ionization mode; HMDB = Human Metabolome Database; HPFS = Health Professionals Follow-Up Study; NHS = Nurses' Health Study; OR = odds ratio; TAG = triacylglycerol; PC = phosphatidylcholine; PE = phosphatidylethanolamine; LPE = lysophosphatidylethanolamine

^a 376 matched case-control pairs.

^b 236 matched case-control pairs.

^c Conditional odds ratios adjusted for family history of CRC, body mass index, smoking history, alcohol consumption (g/d), Alternative Healthy Eating Index, physical activity, menopausal/hormone use status, and regular aspirin use (>2 times/week) in addition to already-matched variables; odds ratio association for 1 SD increase in log-metabolite values.

^d Only the metabolite associations with a P < .05 are shown. The associations from men are shown for comparison.

^e Bold indicates a false discovery rate <0.10.

^f Representative Human Metabolome Database identifier.

Table 4. Association of metabolite modules with risk of CRC and CRC subsites in women (NHS), men (HPFS), and the combined cohort

NHS			All CRC (n = 409)			Proximal (n = 201)			Distal (n = 109)			Rectal (n = 85)		
Module	Metabolites, No.		OR (95% CI) ^a	P ^b	False discovery rate ^c	OR (95% CI) ^a	P ^b	False discovery rate ^c	OR (95% CI) ^a	P ^b	False discovery rate ^c	OR (95% CI) ^a	P ^b	False discovery rate ^c
Light Salmon	27		0.78 (0.67 to 0.92)	.002	0.016	0.83 (0.66 to 1.04)	.110	0.483	0.75 (0.56 to 1.01)	.062	0.431	0.74 (0.49 to 1.11)	.144	0.666
Light Yellow	23		1.11 (0.95 to 1.31)	.190	0.664	1.09 (0.87 to 1.36)	.469	0.656	1.28 (0.88 to 1.85)	.197	0.433	1.09 (0.74 to 1.59)	.671	0.898
Light Sea Green	122		0.92 (0.79 to 1.08)	.319	0.723	0.87 (0.69 to 1.11)	.276	0.483	0.77 (0.55 to 1.08)	.132	0.433	0.98 (0.69 to 1.38)	.898	0.898
Light Slate Blue	12		0.94 (0.80 to 1.10)	.465	0.723	0.84 (0.66 to 1.08)	.169	0.483	1.05 (0.76 to 1.45)	.764	0.764	0.95 (0.67 to 1.35)	.780	0.898
Light Green	22		1.03 (0.90 to 1.18)	.692	0.723	1.00 (0.82 to 1.22)	.990	0.990	1.08 (0.84 to 1.39)	.561	0.654	0.94 (0.61 to 1.44)	.771	0.898
Light Blue	34		1.03 (0.89 to 1.18)	.704	0.723	0.95 (0.77 to 1.17)	.628	0.732	1.18 (0.89 to 1.56)	.247	0.433	1.23 (0.90 to 1.66)	.190	0.666
Light Pink	15		1.03 (0.88 to 1.20)	.723	0.723	1.14 (0.91 to 1.42)	.264	0.483	0.90 (0.65 to 1.25)	.521	0.654	0.85 (0.59 to 1.22)	.379	0.885

HPFS			All CRC (n = 275)			Proximal (n = 104)			Distal (n = 75)			Rectal (n = 66)		
Module	Metabolites, No.		OR (95% CI) ^a	P ^b	False discovery rate ^c	OR (95% CI) ^a	P ^b	False discovery rate ^c	OR (95% CI) ^a	P ^b	False discovery rate ^c	OR (95% CI) ^a	P ^b	False discovery rate ^c
Dark Salmon	28		1.40 (1.10 to 1.79)	.006	0.058	1.30 (0.86 to 1.98)	.208	0.475	1.98 (1.15 to 3.40)	.013	0.060	1.08 (0.62 to 1.88)	.781	0.907
Dark Orange	18		1.19 (0.97 to 1.47)	.102	0.311	1.28 (0.89 to 1.84)	.186	0.475	1.93 (1.21 to 3.06)	.006	0.050	0.83 (0.52 to 1.32)	.432	0.698
Dark Blue	35		1.20 (0.96 to 1.49)	.104	0.311	1.18 (0.85 to 1.64)	.327	0.588	1.19 (0.75 to 1.88)	.452	0.781	1.50 (0.87 to 2.60)	.144	0.698
Dark Magenta	13		1.08 (0.88 to 1.33)	.464	0.853	1.01 (0.72 to 1.41)	.957	0.958	1.14 (0.73 to 1.78)	.578	0.781	1.18 (0.76 to 1.82)	.466	0.698
Dark Red	22		0.95 (0.78 to 1.16)	.633	0.853	0.80 (0.57 to 1.13)	.211	0.475	1.20 (0.81 to 1.77)	.366	0.781	0.99 (0.60 to 1.61)	.953	0.953
Dark Khaki	23		1.05 (0.84 to 1.32)	.645	0.853	0.92 (0.64 to 1.33)	.664	0.958	1.16 (0.75 to 1.79)	.503	0.781	1.07 (0.61 to 1.90)	.806	0.907
Dark Green	65		1.04 (0.86 to 1.27)	.670	0.853	0.99 (0.73 to 1.34)	.958	0.958	0.95 (0.64 to 1.40)	.781	0.781	1.29 (0.79 to 2.12)	.315	0.698
Dark Orchid	13		0.98 (0.81 to 1.18)	.805	0.853	1.04 (0.77 to 1.40)	.789	0.958	1.08 (0.71 to 1.64)	.722	0.781	0.84 (0.55 to 1.31)	.451	0.698
Dark Slate Blue	20		0.98 (0.81 to 1.19)	.853	0.853	1.25 (0.89 to 1.76)	.192	0.475	0.92 (0.64 to 1.30)	.625	0.781	0.77 (0.48 to 1.23)	.271	0.698

Both (consensus)			All CRC (n = 684)			Proximal (n = 305)			Distal (n = 184)			Rectal (n = 151)		
Module	Metabolites, No.		OR (95% CI) ^a	P ^b	False discovery rate ^c	OR (95% CI) ^a	P ^b	False discovery rate ^c	OR (95% CI) ^a	P ^b	False discovery rate ^c	OR (95% CI) ^a	P ^b	False discovery rate ^c
Black	12		1.14 (1.00 to 1.29)	.046	0.283	1.13 (0.93 to 1.36)	.207	0.415	1.52 (1.16 to 1.99)	.002	0.019	0.94 (0.71 to 1.23)	.644	0.818
Brown	18		0.90 (0.80 to 1.01)	.071	0.283	0.95 (0.80 to 1.13)	.578	0.771	0.83 (0.67 to 1.03)	.093	0.248	0.89 (0.69 to 1.14)	.348	0.818
Pink	10		0.92 (0.82 to 1.04)	.182	0.486	0.89 (0.74 to 1.06)	.191	0.415	0.95 (0.76 to 1.20)	.686	0.784	0.97 (0.75 to 1.24)	.793	0.818
Blue	18		0.95 (0.84 to 1.06)	.347	0.657	0.91 (0.76 to 1.09)	.294	0.471	0.80 (0.63 to 1.02)	.068	0.248	1.03 (0.79 to 1.34)	.818	0.818
Turquoise	32		1.04 (0.93 to 1.17)	.476	0.657	1.03 (0.87 to 1.23)	.720	0.823	0.94 (0.75 to 1.17)	.589	0.784	1.04 (0.81 to 1.33)	.782	0.818
Green	16		1.04 (0.93 to 1.17)	.493	0.657	1.17 (0.97 to 1.41)	.100	0.415	0.94 (0.75 to 1.19)	.609	0.784	0.89 (0.70 to 1.14)	.375	0.818
Yellow	16		0.98 (0.87 to 1.09)	.688	0.694	1.00 (0.84 to 1.18)	.974	0.974	0.98 (0.77 to 1.25)	.885	0.885	1.08 (0.83 to 1.41)	.547	0.818
Red	15		1.02 (0.91 to 1.15)	.694	0.694	0.89 (0.75 to 1.06)	.179	0.415	1.10 (0.88 to 1.36)	.404	0.784	1.06 (0.82 to 1.37)	.648	0.818

Abbreviations: CRC = colorectal cancer; HPFS = Health Professionals Follow-Up Study; NHS = Nurses' Health Study; OR = odds ratio.

^a Conditional odds ratios adjusted for family history of CRC + body mass index, smoking history, alcohol consumption (g/d), Alternative Healthy Eating Index, physical activity, and regular aspirin use (>2 times/week) in addition to already-matched variables, odds ratio association for 1 SD increase in the eigenmetabolite. Menopause/hormone use was also adjusted for in the NHS cohort. Sex (cohort) was adjusted for in both cohorts.^b Bold values indicate a P-value < .05.^c Bold values indicate a false discovery rate < 0.10.

prevents accounting for intraindividual variation. Prior studies from our cohorts, however, had intraclass correlation coefficients for most metabolites between 0.48 and 0.72 over 2 to 3 years.¹³ Second, because the blood samples were shipped using dry ice, there was a delay of approximately 24 hours between collection and processing. We addressed this limitation by excluding metabolites with poor stability during processing. Third, our cohort was mostly White, which may limit the generalizability of our findings. Fourth, due to high correlations between metabolites in the same pathway, reported associations may be linked with another causal metabolite.

In conclusion, we identified CRC-associated prediagnosis metabolites. In addition, we observed that distinct metabolomic profiles were associated with CRC in women compared with men. These results provide insight into the etiopathogenesis of CRC and may have implications for risk prediction and prevention strategies.

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Supplementary material

Supplementary material is available at *JNCI: Journal of the National Cancer Institute* online.

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Conflicts of interest

A.T.C. reports personal fees from Bayer Pharma AG, Pfizer Inc, and Boehringer Ingelheim for work unrelated to this study.

Data availability

Further information, including the procedures to obtain and access data from the NHS and HPFS is described at <https://www.nurseshealthstudy.org/researchers> (contact email: nhsaccess@channing.harvard.edu) and <https://sites.sph.harvard.edu/hpfs/for-collaborators/>.

Data are available upon reasonable written request. According to standard controlled access procedure, applications to use NHS, NHS II, and HPFS resources will be reviewed by our External Collaborators Committee for scientific aims, evaluation of the fit of the data for the proposed methodology, and verification that the proposed use meets the guidelines of the Ethics and Governance Framework and the consent that was provided by the participants. Investigators wishing to use NHS, NHS II, and HPFS data are asked to submit a brief description of the proposed project (go to <https://www.nurseshealthstudy.org/researchers>

(contact email: nhsaccess@channing.harvard.edu) and <https://sites.sph.harvard.edu/hpfs/for-collaborators/> for details).

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